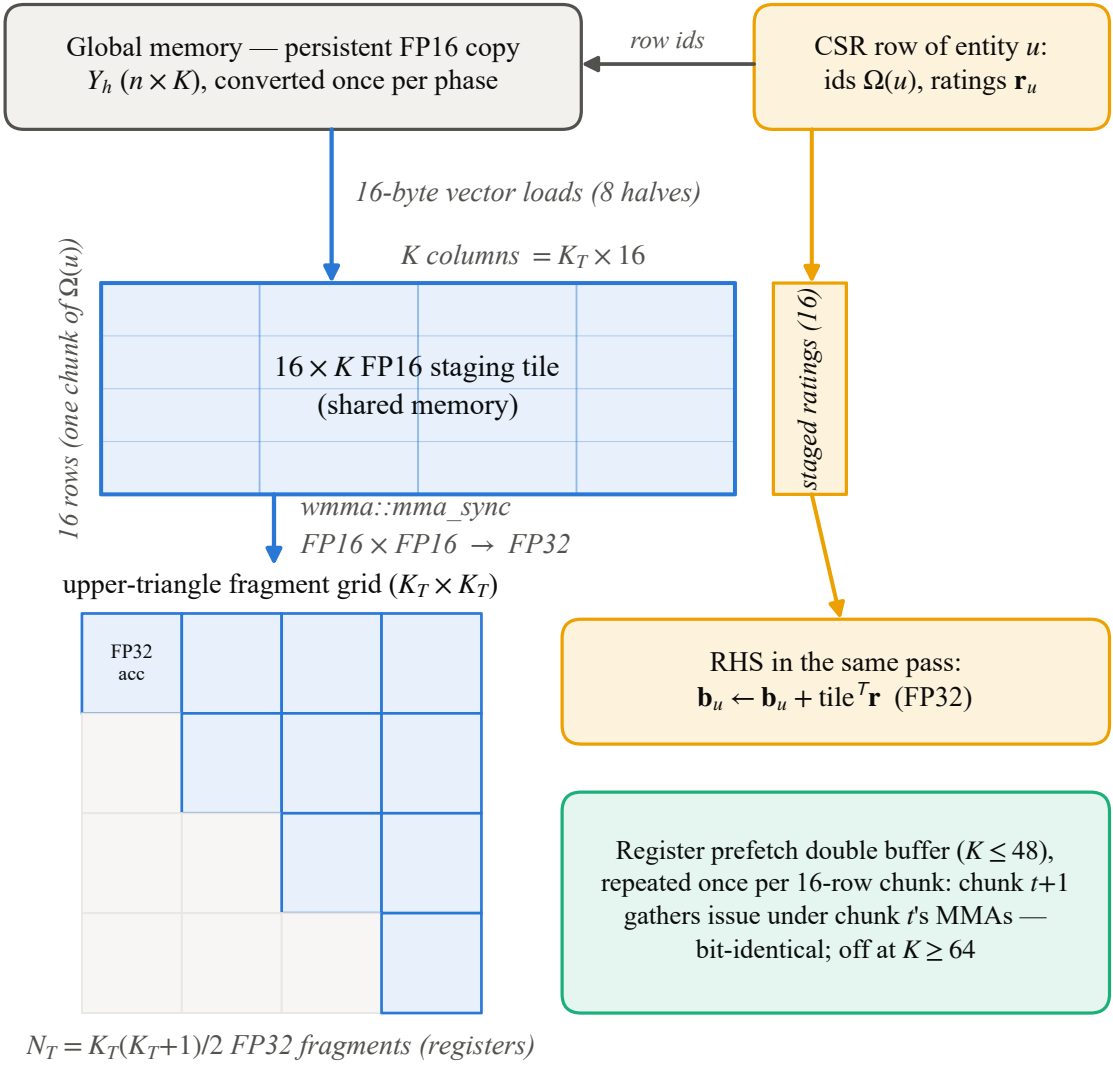

Global memory — persistent FP16 copy
Y_h ($n \times K$), converted once per phase
row ids
CSR row of entity u:
ids $\Omega(u)$, ratings $\mathbf{r}_u$
16-byte vector loads (8 halves)
K columns $= K_T \times 16$
16 rows (one chunk of $\Omega(u)$)
$16 \times K$ FP16 staging tile
(shared memory)
staged ratings (16)
wmma::mma_sync
FP16 × FP16 → FP32
upper-triangle fragment grid ($K_T \times K_T$)
FP32
acc
RHS in the same pass:
$\mathbf{b}_u \leftarrow \mathbf{b}_u + \text{tile}^T\mathbf{r}$ (FP32)
Register prefetch double buffer ($K \le 48$),
repeated once per 16-row chunk: chunk t+1
gathers issue under chunk t's MMAs —
bit-identical; off at $K \ge 64$
$N_T = K_T(K_T+1)/2$ FP32 fragments (registers)